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A predict-then-optimize framework for multi-stage agricultural production and distribution planning

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HIGHLIGHTS

- Machine learning forecasts support agricultural planning decisions.
- Planting, harvesting, and distribution are optimized jointly.
- Flexible harvesting and multi-channel sales improve coordination.
- Prediction-based decisions outperform experience-based planning.

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ABSTRACT

The high uncertainty of crop prices poses significant challenges for decision-making in agricultural production. Farmers often rely on empirical knowledge without access to future market conditions, which can lead to considerable decision-making errors and financial losses. This study introduces a predict-then-optimize (PO) framework that leverages data-driven methods to support farmers in making reliable and accurate production and distribution decisions. A support vector machine algorithm is employed to predict crop prices. The prediction results are then integrated into a multi-stage planning model that focuses on multi-crop cultivation, flexible harvesting, and multi-channel distribution. The model optimizes decisions regarding crop types, planting schedules, cultivation areas, harvest timings, quantities, and distribution channels. Unlike conventional scenario-based methods limited by discrete stages and simplified price uncertainty representations, the proposed framework enables continuous decision-making adaptations across the entire agricultural production and distribution lifecycle. By comparing the efficiency of the PO method with traditional experiential methods, this study demonstrates the advantages of data-driven approaches in enhancing farm profitability. The findings reveal that, regardless of farmers' psychological price expectations, the PO method consistently leads to higher profits.

1. Introduction

In the process of scaled-up agricultural production, new agricultural business has sprung up in several developing countries. For instance, according to data from China's Ministry of Agriculture and Rural Affairs, by the end of 2023, there were over 2.22 million farmer cooperatives and 4 million family farms, which have become the "main force" of modern agriculture [1]. How to develop reasonable production and operation plans for these entities is a crucial issue, which not only affects farmers' incomes but also plays an important role in stabilizing the supply and demand of agri-foods in the market. However, the long production cycles of crops with high price volatility make decision-making difficult for farmers [2]. In most of scenarios, they rely on personal experience and intuition, which can vary greatly and lead to significant decision-making errors [3]. Furthermore, family farms are increasingly adopting multi-crop cultivation and multi-channel distribution practices to manage risks and expand market coverage [4]. This further complicates the relationship between product prices and production management decisions, requiring a more systematic approach to improve farm profitability.

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To address price volatility in decision-making of agricultural production planning, some studies have modeled price uncertainty under numerable scenarios, typically using methods such as Monte Carlo simulations or approximate distributions [5,6]. These methods usually generate a certain number of high, medium, and low-price scenarios, and chose a best plan to maximize the expected profits (or minimize the expected costs) among all scenarios. In this case, a small number of decision stages is crucial to avoid the "curse of dimensionality". However, the production planning of crops often needs continuous decision-making among multiple weeks [7], especially under the practice of multi-crop cultivation, flexible harvesting, and multi-channel distribution. The scenario-based methods are therefore not suitable to address the problem at hand. Moreover, in such settings, production and distribution decisions are closely interconnected, since planting and harvest arrangements directly determine downstream sales opportunities across different channels.

This paper adopts a predict-then-optimize (PO) framework to address the impact of uncertain prices on multi-stage production planning, with the aim of providing a robust tool that enhances the accuracy and reliability of farmers' decisions. A support vector machine algorithm is utilized to predict the multi-period price changes of crops in the future based on historical prices. The predicted prices are then integrated into a multi-stage planning model that is proposed for family farms, with the consideration of multi-crop cultivation, flexible harvesting, and multi-channel distribution. Following decisions can be obtained from the model: 1) Crop types, planting schedules, and cultivation areas; 2) Harvest timing and quantities for different crops; 3) Distribution channels and sales volumes for various crops. This predict-based approach can reduce farmers' reliance on experience and intuition, enabling farmers to better respond to market volatility. However, its effectiveness in improving farm profitability remains uncertain. Thus, we compare the PO approach with simulated experiential decisions made by farmers based on prospect theory to examine the efficiency differences between experiential and predict-based decisions under various scenarios. We find that, regardless of farmers' psychological price expectations, the PO method consistently leads to higher farm profitability.

The contributions of this paper mainly focus on the following three aspects: (1) It constructs a PO framework that integrates price forecasting with multi-stage agricultural production and distribution planning, which extends the application of PO methods to scenarios of continuous decision-making under price fluctuations. (2) It proposes a joint optimization model that simultaneously considers multi-crop plantings, flexible harvesting, and multi-channel distribution, thereby capturing the interdependence among farmers' key production and distribution decisions. (3) It compares prediction-based decision-making with experience-based decision-making under different price expectation scenarios, revealing the performance differences between the two in agricultural PDP.

The remainder of this paper is organized as follows: Section 2 reviews the related work. Section 3 presents the PO approach for crop production planning, which includes an introduction to the PO framework, a crop price predict approach, and a multi-stage planning model. Section 4 offers the results and discussion from a case study application. Finally, Section 5 summarizes this study and provides future research directions.

2. Literature review

We can classify the relevant studies into two categories. First, studies that focused on the production and distribution planning (PDP) of crops, including both traditional PDP models and their variations. Second, studies that delved into the application of the PO framework across various fields with an emphasis on supply chain planning.

2.1. Production and distribution planning of crops

The PDP issue was first raised by Glover et al. [8] in 1979. It derives from manufacture industry [9], aiming to achieve coordination of production and distribution to better reduce total operating costs, and has since been extended to the agricultural sector [10]. The PDP problem for crops involves complex decisions regarding planting, cultivation, harvesting, storage, and distribution, aiming to minimize product losses and operating costs while maximizing profitability [11]. A seminal contribution to this field is the study by Glen [12], which applied operations research to address agricultural planning issues in both crop and livestock sectors and proposed corresponding production planning models. This study laid a foundation for subsequent research by demonstrating the effectiveness of mathematical modeling in addressing complex agri-food planning challenges. Another representative study is that of Ahumada and Villalobos [13], which expanded traditional models by incorporating postharvest planning considerations. This research underscored the importance of considering post-harvest processes, such as storage and preservation, in reducing food wastage and improving supply chain efficiency.

Given the multi-stage nature of crop production and distribution, existing studies differ in their system structures and modeling approaches. Some studies primarily focus on upstream planting decisions, without explicitly considering downstream harvesting and distribution. Others concentrate on planting-harvesting system, analyzing the dynamic changes in harvest timing, crop maturity, and labor allocation. Still others extend the system boundaries to harvesting-distribution systems or integrated production-distribution systems, incorporating downstream logistics arrangements and channel allocation into a unified analytical framework. The time organization of these studies also differs: some assume a relatively fixed planting-to-harvest cycle, while others employ multi-period or rolling-horizon structures to capture sequential adjustments over time. These differences directly impact the ability of these models to capture the interdependencies between planting, harvesting, and distribution decisions in actual agricultural production.

For example, some studies have focused on crop planning, such as the work by Otoo et al. [14], who developed a linear programming model for optimizing crop selection to maximize profits of small-scale farmers in Ghana. Gómez-Lagos et al. [15] concentrated on harvest planning, with an aim to optimize harvest decisions and minimize total costs in fruit orchards. Other research has integrated technological advancements like the Internet of Things (IoT) to enhance distribution planning. For example, Ruan et al. [16] investigated the application of IoT sensors for real-time monitoring of fruit freshness, thereby enabling more responsive and precise agricultural practices. Similarly, Jiang et al. [17] proposed an integrated fresh fruit harvest and distribution model that quantifies post-harvest maturity using fuzzy classification and kernel clustering. By incorporating maturity into a time–space network optimization framework, their approach improved service quality by up to 50.21% compared with traditional models, demonstrating the benefits of jointly considering quality attributes and logistics efficiency.

Certain studies have delved into multi-category cultivation or multi-channel sales strategies. For example, Sarımehmet et al. [18] addressed the tea harvest scheduling problem by considering both factory capacity and farmers' harvest day preferences. They developed mathematical models that not only minimized factory expeditions but also significantly improved farmers' compliance with preferred harvest days—from 52% to 97%—without negatively affecting factory efficiency. Sisley et al. proposed a mixed integer programming (MIP) model for managing the production of multiple crops within a multi-planted area, ensuring a consistent supply of fresh agri-food without storage by adjusting production plans over an annual time frame [19]. Han et al. divided farmers' distribution channels into contract, e-commerce, wholesale, and retail, and developed a MIP model to help farmers optimize their production and distribution plans based on price and

demand differences across these channels to maximize income [20]. Recently, there is an increasing emphasis on integrating sustainable practices into production and distribution planning. For example, Mishra et al. explored the impact of sustainable farming techniques on overall supply chain efficiency, offering insights into the trade-offs between environmental sustainability and economic viability [21]. It can be seen that the increasingly complex nature of modern agriculture presents new challenges to the study of PDP issues, requiring models capable of addressing complex decision-making processes in the context of climate change, new business models, and technological innovations.

Price volatility is a critical factor that impacts crop production planning, which is often addressed based on stochastic modeling methods in existing studies. Ahumada et al. [22] tackled price uncertainty of fresh produce by fitting historical price data to lognormal distributions, using multiplicative seasonal models, and incorporating autoregressive components. Their methods generate finite price scenarios that capture seasonal patterns and autocorrelations. Flores and Villalobos [23] addressed price uncertainty by transforming weekly price data into a multivariate normal distribution, despite observed deviations from normality. This approach, while acknowledging the violations of normality assumptions, allowed for the generation of representative weekly price values with satisfactory skewness and heavy tails, providing a practical method for capturing annual price variability. Nadal-Roig et al. [5] addressed price uncertainty in vertically integrated pig production using a scenario tree based on Monte Carlo sampling. They generated 27 distinct price scenarios from historical data, enabling better planning for piglet purchases and farm rentals. Similar to their studies, Alemany et al. [11] addressed the challenge of price uncertainty in crop planning by representing uncertain parameters as triangular fuzzy numbers. The most possible value for each parameter is based on deterministic estimates, with pessimistic and optimistic values determined by fixed percentage adjustments. While these models provide valuable insights, they often struggle to capture all possible scenarios, limiting their application to long-term planning.

In addition to price uncertainty, other forms of operational uncertainty have been examined in recent work. Komijani and Sajadieh [24] developed an integrated production–inventory–routing model for a green closed-loop supply chain that explicitly incorporates stochastic lifespans of perishable goods. Their results show that ignoring lifespan variability can underestimate producer costs by as much as 25%, highlighting the risks of oversimplifying perishability in planning models. Likewise, Gámez-Albán et al. [25] investigated the “first mile” in agri-food supply chains and proposed a mixed-integer programming approach using a heterogeneous fleet. They demonstrated that reorganizing pick-up rounds with cooled vehicles can reduce logistics costs by up to 40%.

Existing research provides important insights into decision-making at different stages of agricultural production and distribution systems. However, integration forms in the literature differ in their decision-making focus and system boundaries: some emphasize production and processing coordination, while others focus on the planting-distribution or harvesting-distribution stages. In contrast, PDP is particularly critical for fresh agricultural products, as upstream crop selection, planting arrangements, and harvest timing directly impact downstream channel allocation, sales opportunities, and profitability. This interdependence between upstream and downstream stages becomes even more pronounced when farms further adopt multi-crop planting, flexible harvesting, and multi-channel distribution strategies. Nevertheless, many existing studies only cover parts of the planting, harvesting and distribution chain or characterize uncertainty using only a limited number of scenarios. In contrast, this paper develops a three-stage planning model for agricultural production and distribution that explicitly links planting, harvesting, and distribution decisions and embeds predicted prices into the optimization model to address the limitations of limited-scenario representations of uncertainty.

2.2. Predict-then-optimize in supply chain planning

The PO framework has been applied to many fields. In this section, we mainly focus on its application study in the context of supply chain planning. For production planning issues, Shiue et al. [26] introduced a reinforcement learning (RL) based real-time scheduling approach that integrates a multiple dispatching rules mechanism to dynamically optimize production performance in a smart factory environment. The study uses an offline learning module in combination with Q-learning RL to predict and adapt scheduling strategies in response to changing shop floor conditions. Wang et al. [27] presented a smart “predict-then-optimize” (SPO) approach for refinery planning within uncertain product prices, integrating temporal graph convolution networks to enhance price prediction by capturing spatio-temporal features. The SPO framework combines the stages of price prediction and refinery planning optimization, using a combined loss function to improve decision-making performance. This method demonstrated significant improvements in both multi-period and single-period refinery planning scenarios by addressing price uncertainty more effectively than traditional approaches. In terms of transportation planning, Basso et al. [28] addressed the electric vehicle routing problem by integrating a PO approach. They used a probabilistic Bayesian machine learning method to predict energy consumption and its variance, allowing for robust route planning under uncertainty. The study varified that considering energy demand within a confidence interval could lead to more reliable and efficient routing solutions compared to deterministic models. In the context of inventory management, Cheung et al. [29] addressed the joint replenishment problem (JRP). They predicted the impact of ordering decisions on total costs and developed approximation algorithms to efficiently minimize these costs while satisfying time-dependent demands, allowing for more accurate and practical solutions for JRP issues. Sharifnia et al. [30] presented a robust simulation optimization approach for supply chain management under uncertainty, employing artificial neural networks (ANNs) as metamodels to predict and optimize safety stock levels. By integrating ANN-based predictions with a Taguchi robust optimization framework, this study efficiently enhances the robustness of supply chain decisions.

In the field of agriculture, Liu et al. [31] developed a PO method integrating the maturity of crops to enhance distribution efficiency. A regression modeling approach was employed to predict maturity changes based on temperature, storage duration, and ethephon concentration. A MIP model was then used to optimize distribution routes with the consideration of predicted product maturity, ensuring timely delivery and minimizing waste and logistics losses. Zheng et al. [32] developed a high-precision predictive model that used an ANN to forecast emissions from a diesel engine running on biodiesel-diesel blends under various conditions. They then applied a covariance matrix adaptation-evolutionary strategy to optimize engine parameters for minimizing emissions during crop harvesting. Results validated the effectiveness of this combined predict and optimize approach. Fan et al. [33] utilized the predicted monthly prices of various crops to make cooperative planting and harvesting decisions. Their findings showed that planting and harvesting plans based on the predicted prices of crops enable better adaptation to fluctuations in market prices. However, it is noteworthy that their research did not take into account decision-making in the distribution stage. Furthermore, they did not compare the optimal results of predict-based approach with those obtained from traditional experience-based method. This introduces a potential limitation, as demonstrated by Elmachtoub et al. [34], who highlighted that accurate parameter predictions do not necessarily imply optimality in decision-making.

2.3. Research Gap Analysis

From the literature reviewed, two critical gaps have been identified: While a wealth of research has explored PDP of crops, existing

studies largely address uncertainty through simplified stochastic models or a limited number of predefined scenarios. These methods are better suited to scenarios with fewer decision-making stages and relatively stable structures, but struggle to characterize the multi-stage decision-making processes that continue for months in crop production. Since the profitability of planting, harvesting, and distribution decisions largely depends on future market prices, existing scenario-based PDP research remains inadequate for addressing multi-stage planning problems under persistent price fluctuations.

Existing research has paid limited attention to the performance differences between decisions based on predicted prices and those based on farmers' experience. This issue is important because improved forecasting accuracy does not necessarily lead to better decision outcomes. In practice, farmers often form future price expectations based on experience or historical prices and make PDP accordingly. Without a direct comparison between these two decision approaches, it remains unclear whether price forecasting can truly improve decision performance in agricultural PDP.

To fill these research gaps, this paper constructs a PO framework for multi-stage agricultural PDP. Specifically, it incorporates predicted prices into a joint optimization model, simultaneously considering decisions such as multi-crop planting, flexible harvesting, and multi-channel distribution. This paper also compares the performance of prediction-based and experience-based decisions under different expectation scenarios in agricultural production and distribution planning.

3. Problem description

This section introduces the proposed PO approach for crop production planning. It starts with an overview of the framework of the PO approach, followed by a discussion of the price predict method and the multi-sage planning model.

3.1. The framework of the predict-then-optimize

The predict-then-optimize framework combines predictive analytics with optimization techniques to support informed decision-making in agriculture under uncertain market conditions. As shown in Fig. 1, the framework starts by collecting relevant input data, including historical prices, market trends, and external factors like weather conditions. These inputs may vary depending on whether a univariate or multivariate prediction model is being used. The predictive model then forecasts future prices, which are not necessarily discrete price scenarios (such as high, medium, or low), but rather continuous price predictions over a specified time horizon, reflecting the dynamic nature of market fluctuations.

The predicted prices are used as input parameters for the optimization model, which recommends the most profitable decisions for production planning, such as planting schedules, harvesting times, and distribution strategies. By incorporating future price trends, the model helps farmers manage risk and maximize profitability despite market volatility.

The framework also incorporates a feedback mechanism, though it is not necessarily an automatic adjustment process. While data can be continuously updated, the adjustments to prediction models and subsequent decisions may be made manually based on the observed effectiveness of previous decisions. This continuous improvement cycle allows farmers to refine both the forecasting and the operational planning models as new data and outcomes become available, enhancing decision accuracy and adaptability over time.

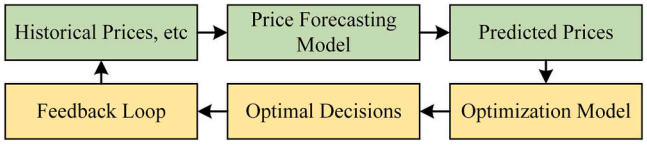


Fig. 1. Design of the predict-then-optimize framework for crop production planning.

The choice of a mathematical programming approach in this study is driven by the structured and moderately sized nature of the target problem. The production and distribution planning tasks for family farms, which involve a defined set of crops, sales channels, and decision periods, can be explicitly formulated as a multi-stage optimization model with well-defined constraints and an objective function representing profit maximization. This structure allows modern optimization solvers to guarantee globally optimal solutions within seconds on standard computing equipment, ensuring both computational efficiency and solution accuracy. While metaheuristic algorithms are valuable for tackling very large-scale or highly unstructured problems, they typically produce near-optimal solutions and may compromise interpretability. This is particularly important in agricultural decision-making, where transparency and reliability are essential for adoption by farmers.

In addition, the proposed predict-then-optimize framework integrates a support vector machine–based price prediction module with the mathematical programming model, combining the adaptability of machine learning in capturing market dynamics with the precision of optimization in resource allocation. This hybrid design enables continuous decision-making adjustments across the entire production and distribution cycle, accommodating features such as multi-crop cultivation, flexible harvesting, and multi-channel distribution. As demonstrated in our case study, the model remains computationally tractable for small to medium-scale family farms, making it a practical and implementable tool for improving decision quality in real agricultural contexts.

3.2. Analysis and prediction of crop prices

In this study, we focus on the broader category of crops; however, for both the price analysis and the optimization model, we specifically use vegetable production and prices as representative proxies. Similar to most crops, vegetable prices show high volatility, perishability, and seasonality. The insights gained from vegetable production can thus be reasonably extended to other types of crops.

Vegetable prices are usually not subject to government-imposed minimum price controls and exhibit a high degree of marketization. Fig. 2 presents the weekly prices of tomatoes, peppers and cucumbers from September 2020 to September 2022, as reported by the Agricultural Products Commercial Information Public Service Platform of the Ministry of Commerce of China. By analyzing the price fluctuations of three types of vegetables, significant differences in prices can be observed both among different vegetables varieties and within the same varieties across various years. For example, the price of chili peppers

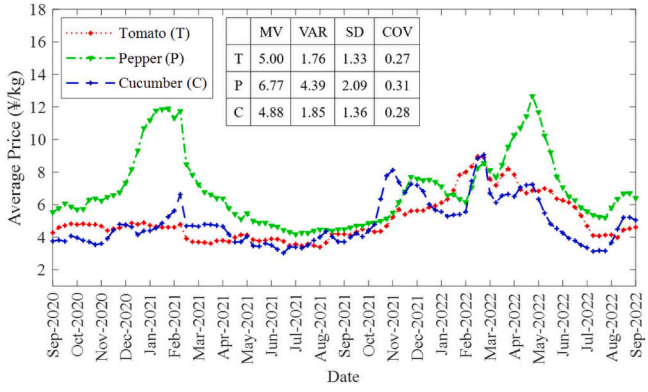


Fig. 2. Average weekly market price of different vegetables (MV: mean value, VAR: variance, SD: standard deviation, COV: coefficient of variation).

reached its first peak in January 2021; however, it experienced an unusual decline in January 2022 and did not reach another peak until May 2022. It can be seen from Fig. 2 that the mean value, variance, standard deviation and coefficient of variation of tomato and cucumber are close to each other, while the difference for pepper is great. Moreover, the coefficient of variation for the prices of these three types of vegetables is close to 30%, indicating a high degree of price variability. This variability in vegetable prices and the significant difference among various vegetable prices presents great challenges to farmers' planting and distribution decisions.

The role of vegetable price prediction in this study is to provide relatively reliable input data for the optimization model. When the values of key parameters are closer to real-world variations, the optimization results of the model are likely to be more reliable. There are various methods for price prediction, mainly categorized into statistical regression and machine learning techniques. Among these, the SVM method is particularly effective in handling small-sample datasets, mitigating noise and outlier impacts, and reducing the risk of overfitting [35]. Therefore, this study adopts the SVM method for vegetable price prediction. The weekly price data from May 16, 2014 to September 6, 2019, serves as the training set, while data from September 13, 2019 to March 20, 2020 is used as the testing set.

We constructed support vector machine prediction models with and without seasonal components. Table 1 presents the mean absolute error (MAE) and R-squared values for each crop's test set predictions under both settings. The results show that removing seasonality improves the prediction performance for all three crops: the MAE values for tomatoes, peppers, and cucumbers are all below 0.3, while the corresponding R-squared values are close to 1. This indicates that filtering out seasonality can more effectively capture short-term price fluctuations, thus providing a more reliable input for the subsequent optimization model.

Fig. 3 illustrates the predicted weekly prices and actual prices of these crops using the SVM method after eliminating seasonal factors, indicating the effectiveness of the employed price predict method.

3.3. Decision support model formulation

To facilitate analysis, we take the vegetable management process of a family farm as an example for modeling. Fig. 4 illustrates the three-stage planning model that involves cropping, harvesting, and distribution. Family farms first need to make planting decisions, mainly including the types of vegetables to be planted, planting times, and planting area. It is important to note that these decisions are highly dependent on future market price information. However, there is a long-time interval between the planting and harvesting of crops. Farmers need to anticipate the price of vegetables at the time of distribution when making their planting decisions.

Following farmers' planting decisions based on market expectations, the next important stage is harvesting. The harvesting time for crops allow for a certain degree of flexibility, which is related to the perishability of vegetables [36]. For example, if the shelf life of a certain vegetable after reaching its physiological maturity period is 15 days, the farmer can choose to harvest it on any day within 15 days. Therefore, farmers' harvesting decisions are made based on the characteristics of vegetable ripeness and perishability.

Farmers sell vegetable products through various distribution channels. The sales prices, costs, and demands of various vegetables vary across different channels. For example, the prices on fresh e-commerce are relatively considerable, but there are also high packaging and logistics costs. The price of contract farming is relatively stable, but the demand is limited. Therefore, farmers need to determine the channels for selling vegetables and determine the sales volume of each channel.

Based on the above analysis, the present question focuses on how farmers develop vegetable planting, harvesting, and distribution plans based on market price expectations. This paper proposes a three-stage mathematical programming model to formulate the complex planning problem faced by family farms. Importantly, the model allows for numerical analysis to obtain meaningful managerial insights. To provide a more intuitive representation of the optimization model framework, we present a textual description below.

Maximize profit = [sales revenue - planting costs- packaging and transport costs - labor costs]

Subject to:

Land area constraints

Labor resource constraints

Capital constraints

Limitations on breach of contract

Picked and unpicked quantity balance conditions

Demand constraints

Let $J = \{1, 2, \ldots n\}$ be the set of feasible planting weeks, where n denotes the number of total weeks in the planting period. Let $H = \{1, 2, \ldots v\}$ be the set of feasible distribution weeks, where v denotes the number of total weeks in the distribution period. Let j and h be the index of planting and distribution weeks, respectively, $j \in J$, $h \in H$. We also consider a set of crops denoted as U, where u represents the index of crop types, $u \in U$.

The symbols and definitions of the decision variables and parameters used in the model are presented in Table 2

The objective function is to maximize farmers' profits, equaling to sales revenue minus planting costs, distribution costs, and labor costs. Eq. (2) indicates that the total planting area of crops during the entire planning period should not exceed the available land area. Eq. (3) ensures that the labor demand generated during the harvesting and distribution does not exceed the labor available to farmers. Constraint (4) indicates that the cost incurred throughout the planning period does not exceed the maximum funds available to farmers. Eq. (5) shows that the quantity of vegetables provided by farmers to agricultural enterprises does not violate the terms of the order contract. Eq. (6) ensures that the number of vegetables picked each week after maturity equals the sales quantity. Eq. (7) limits the quantitative relationship between each week's unpicked, ripe, and harvested crop. Eq. (8) indicates that the distribution quantity of each product on the e-commerce platform should be less than the expected demand of the e-commerce platform. Constraint (9) indicates that the retail amount of each product should be less than the expected total sales volume of retail outlets. Eqs. (10) - (11) ensure that the sales volume in week h is greater than the unpicked quantity of the last week and less than the marketable quantity of this week. Eq. (12) limits the domain of decision variables.

$$\begin{aligned}\text{MaximizeF} = & \sum_{h\in H}\sum_{u\in U}\left(p^c_{hu}x^c_{hu} + p^e_{hu}x^e_{hu} + p^w_{hu}x^w_{hu} + p^r_{hu}x^r_{hu}\right) - \sum_{j\in J}\sum_{u\in U}\text{cpl}_u q_{ju} - \sum_{h\in H}\sum_{u\in U}\frac{\left(ct^e + cp^e\right)x^e_{hu}}{w^e} \\ & - cl\left[\sum_{j\in J}\sum_{h\in H}\sum_{u\in U} q_{ju}\left(\frac{M_{jhu}}{ml_u} + \frac{R_{jhu}ay_u}{rl_u}\right) + \sum_{h\in H}\sum_{u\in U}\left(\frac{x^e_{hu}t^e}{w^e lt}\right) + \sum_{h\in H} bq^r_h + \sum_{h\in H} l^e_h\right)\right]\end{aligned} \tag{1}$$

Table 1
Comparison of MAEs and R-Squared of various crops using the SVM method.

Crop	Tomato	Pepper	Cucumber	Seasonal components
MAE_1	0.24	0.26	0.25	Stripping off
R-Squared_1	0.91	0.96	0.87	
MAE_2	0.31	0.26	0.31	Containing
R-Squared_2	0.87	0.95	0.83	

s.t.

$$\sum_{j\in J}\sum_{u\in U} q_{ju} \le fs \tag{2}$$

$$\sum_{j\in J}\sum_{u\in U} q_{ju}\left(\frac{M_{jhu}}{ml_u}+\frac{R_{jhu}ay_u}{rl}\right)+\sum_{u\in U}\left(\frac{x^e_{hu}t^e}{w^e lt}\right)+bq^r_h+l^e_h \le la_h, \forall h\in H \tag{3}$$

$$\sum_{j\in J}\sum_{u\in U} q_{ju}\mathrm{cpl}_u+\sum_{h\in H}\sum_{u\in U}\frac{(ct^e+cp^e)x^e_{hu}}{w^e} +cl\left[\sum_{j\in J}\sum_{h\in H}\sum_{u\in U} q_{ju}\left(\frac{M_{jhu}}{ml_u}+\frac{R_{jhu}ay_u}{rl_u}\right)+\sum_{h\in H}\sum_{u\in U}\left(\frac{x^e_{hu}t^e}{w^e lt}\right)+\sum_{h\in H} bq^r_h+\sum_{h\in H} l^e_h\right]\le fc \tag{4}$$

$$oq^c\sum_{j\in J} q_{ju}S_{jhu}ay_u \le x^c_{hu}, \forall h\in H, u\in U \tag{5}$$

$$qh_{hu}=\sum_{u\in U}\left(x^c_{hu}+x^e_{hu}+x^w_{hu}+x^r_{hu}\right), \forall h\in H \tag{6}$$

$$nh_{hu}\begin{cases}\sum_{j\in J} ay_u q_{ju}S_{jhu}-qh_{hu}, h=1\\ \sum_{j\in J} ay_u q_{ju}S_{jhu}-qh_{hu}+nh_{h-1,u}, 2\le h<v\\ nh_{hu}=0, h=v\end{cases} \tag{7}$$

$$\sum_{h\in H} x^e_{hu} \le f^e_u, \forall u\in U \tag{8}$$

$$\sum_{u\in U} x^r_{hu} \le mq^r bq^r_h, \forall h\in H \tag{9}$$

$$\left(x^c_{hu}+x^e_{hu}+x^w_{hu}+x^r_{hu}\right)\le \sum_{j\in J} ay_u q_{ju}S_{jhu}, \forall u\in U, h=1 \tag{10}$$

$$nh_{h-1,u}\le\left(x^c_{hu}+x^e_{hu}+x^w_{hu}+x^r_{hu}\right)\le nh_{h-1,u}+\sum_{j\in J} ay_u q_{ju}S_{jhu}, \forall u\in U, 2\le h\le v \tag{11}$$

$$q_{ju}\ge 0, x^c_{hu}\ge 0, x^e_{hu}\ge 0, x^w_{hu}\ge 0, x^r_{hu}\ge 0, qh_{hu}\ge 0, nh_{hu}\ge 0 \tag{12}$$

The proposed model is computationally tractable for small to medium scale family farm operations. In the present case study, it simultaneously optimizes production and distribution decisions for three crops, four sales channels, and a 20-week planning horizon, resulting in 306 decision variables and 660 constraints after formulation. The model was implemented in MATLAB using the YALMIP toolbox and solved with the Gurobi Optimizer version 10.0.3 on a standard desktop computer equipped with an AMD Ryzen 5 3550 H CPU with four physical cores and eight logical processors, and 16 GB of RAM. The solution process

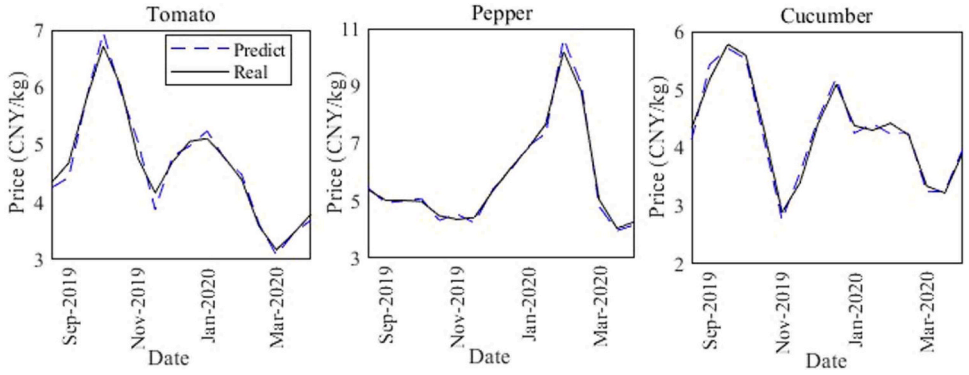


Fig. 3. The real and the predicted weekly prices for each crop.

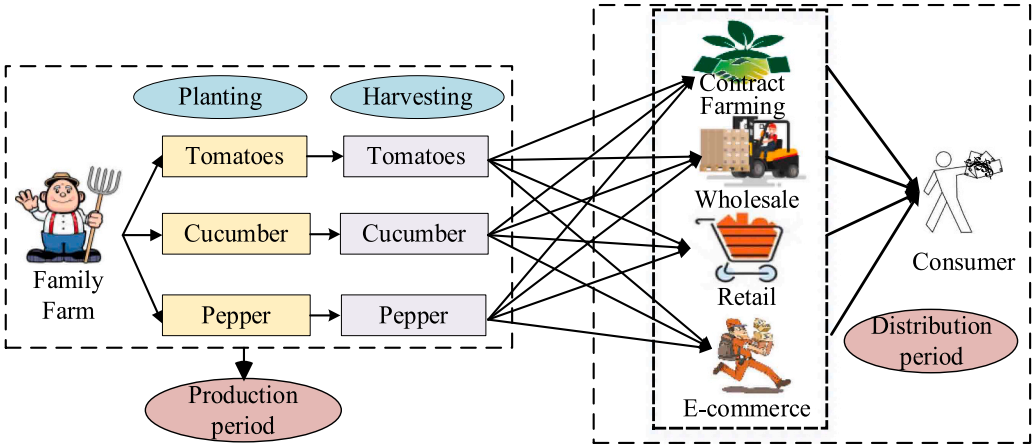


Fig. 4. Illustration of the three-stage planning model.

Table 2
Decision variables and parameters.

Symbols	Definition
parameters	
p^c_{hu}	Price of crop u on contract farming in period h
p^e_{hu}	Price of crop u on fresh e-commerce in period h
p^w_{hu}	Price of crop u on wholesale in period h
p^r_{hu}	Price of crop u on retail in period h
cpl_u	Planting cost of crop u (ha)
ct^e	Express delivery cost for fresh e-commerce
cp^e	Cost to package an e-commerce parcel
cl	Cost of laborers per man-week
ml_u	The area of crop u that each laborer can manage.
M_{jhu}	Equal to 1 when crop u is planting at period j and cultivating at period h, otherwise equal to 0
R_{jhu}	Equal to 1 when crop u is planting at period j and harvesting at period h, otherwise equal to 0
ay_u	Total yield of crop u (kg/ha)
rl_u	Weight of crop u that a worker can harvest in a week
t^e	The time required to pack an e-commerce package
w^e	Net weight of an e-commerce package
lt	Number of working hours per week for workers
mq^r	The maximum expected demand for a single retail stall.
bq^r_h	Number of retail outlets at period h
l^e_h	Number of laborers required for e-commerce operation in period h
fs	Land available for farmers (ha)
f^e_u	Expected maximum sales volume of crop u on e-commerce platform
s_{jhu}	Proportion of crop u planted at week j and harvested in week h to the total yield
la_h	Number of labor available at period h
fc	Available funds during the planning period
oq^c	Guaranteed supply percentage
Decision variables	
q_{ju}	Area to plant crop u in period j
qh_{hu}	Picking quantity of crop u in period h
nh_{hu}	Unpicked quantity of crop u in period h
x^c_{hu}	Sales quantity of crop u in contract farming in period h
x^e_{hu}	Sales quantity of crop u in fresh e-commerce in period h
x^w_{hu}	Sales quantity of crop u in wholesale in period h
x^r_{hu}	Sales quantity of crop u in retail in period h

involved 156 iterations and was completed in 0.07 s. These results indicate that the model can be solved efficiently within practical time limits, making it suitable for real-world agricultural decision-making scenarios of similar scale.

Compared to existing agricultural planning models, the novelty of the proposed mathematical model lies in integrating planting, harvesting, and distribution decisions into a unified three-stage optimization structure. This enables the model to capture the dependency between upstream production decisions and downstream distribution decisions under price uncertainty. The model also considers the flexibility of harvest time, agricultural products may be harvested within a feasible maturity window, but are constrained by perishability. This allows the model to optimize harvest timing and quantity under the combined influence of price, demand, and labor constraints, which helps reduce losses caused by poorly timed harvesting. The model clearly distinguishes four distribution channels, namely contract farming, fresh e-commerce, wholesale, and retail. This channel-specific design more closely reflects the multi-channel sales environment faced by farmers in practice and facilitates coordinated allocation among different channels to spread market risk and stabilize farmers' income.

4. Results and discussion

In this section, we present the results and engage in a discussion of our findings. We start by analyzing the base case scenario, focusing on planting schedules, harvesting decisions, and distribution outcomes. Next, we compare the difference in farmers' decision-making results under the influence of predicted prices and expected prices. We then explore the influence of key factors on farm profitability, including multi-crop cultivation, labor efficiency, harvesting flexibility, and available funds.

Table 3
Detailed planting schedule calculated by the decision-support model.

Crop	Planting weeks						
	1	2	3	4	5	6	Total
Tomato	0.03	0.07	0.32	0.00	0.63	0.41	1.46
Pepper	0.00	0.00	0.00	0.00	0.65	1.44	2.09
Cucumber	0.03	0.21	0.00	1.21	0.00	0.00	1.45
Total	0.06	0.92	0.32	1.21	1.28	1.85	5.00

4.1. Results of the base case

The numerical analysis is based on a case study of a family-owned greenhouse vegetable farm in Jingyang County, Shaanxi Province, China. The farm mainly grows vegetables, particularly tomatoes, peppers, and cucumbers. The data used in this case study come from various sources. Wholesale vegetable prices and online demand are sourced from the Agricultural Business Information Platform of the Ministry of Commerce of China and JD.com. Contract farming prices are obtained from Han et al. [37]. Various vegetable production costs are derived from the 2021 Chinese Agricultural Cost and Revenue Data Compilation. Changes in yield after vegetable maturity (Table 3) are taken from Flores and Villalobos [23] and Ahumada and Villalobos [13]. The available labor force is based on micro-research data collected through local field research by our team.

The model proposed in this paper was encoded using YALMIP and solved with the Gurobi solver. The model was run on an AMD Ryzen 5, 2.10 GHz processor with 16 GB of RAM.

4.1.1. Planting decisions

Crop cultivation is the first step in vegetable production. A planting plan should include crop variety, planting time, and planting scale. When a family farm has 5 ha of land, the planting plan generated by the model is shown in Table 3. In this scenario, farmers should plant all three crops. Tomatoes and chili peppers should be planted mainly in the early stages, with a lighter workload. For example, 0.07 ha of tomatoes and 0.21 ha of cucumbers should be planted in the second week. The workload increases significantly in the middle and later stages, with 1.21, 1.28, and 1.85 ha to be planted in weeks 4, 5, and 6, respectively.

4.1.2. Harvesting and distribution decisions

Determining the appropriate harvesting time is also a key decision that farmers need to make. Fig. 5(a) shows a detailed picking plan for the three types of vegetables, with the picking work mainly focused on weeks 4–14. For example, in week 6, approximately 17,500 kg of chili peppers, 8000 kg of tomatoes, and 2500 kg of cucumbers must be picked.

After crop harvesting, farmers need to make distribution decisions. As shown in Fig. 5(b) ~ (e), detailed distribution plans for tomatoes, peppers, and cucumbers are presented through four sales channels: contract farming, fresh e-commerce, wholesale, and retail. For example, when the farm has 5 ha of arable land, a total of 109,000 kg of vegetables are sold through order farming, with tomatoes, chili peppers, and cucumbers accounting for 29.79%, 35.01%, and 35.21% of the sales volume, respectively. In the 12th week, farmers harvest vegetables according to the harvest schedule and sell about 16,000 kg of peppers on e-commerce platforms. Based on the distribution plan, it can also be known that selling about 9000 kg of cucumbers through retail channels is most beneficial for farmers. These results indicate that the proposed model coordinates planting, harvesting, and distribution decisions jointly rather than optimizing each stage in isolation. This coordination helps explain why prediction-based planning performs better in the subsequent analysis.

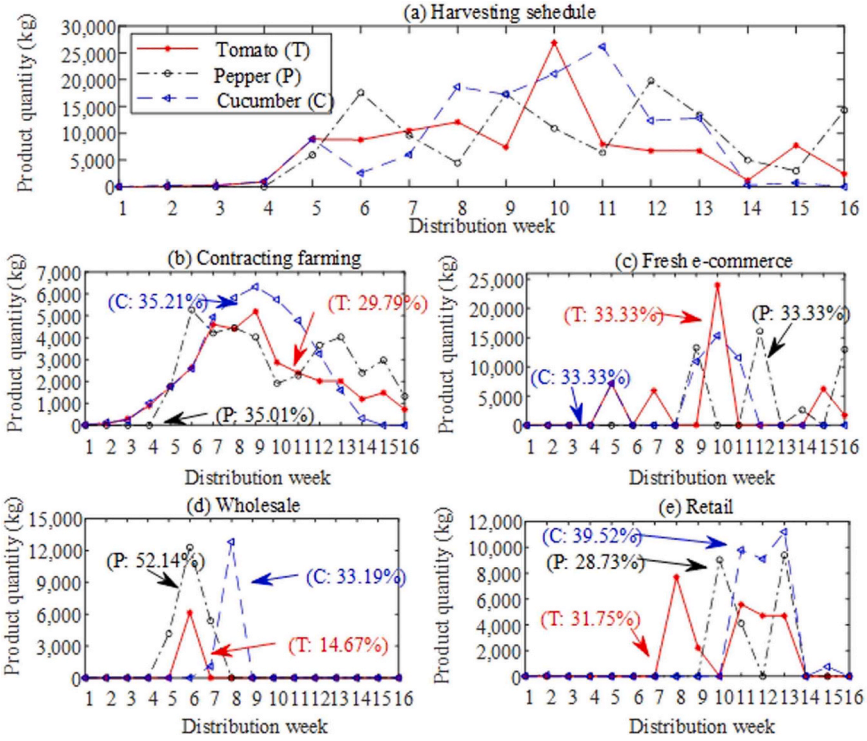


Fig. 5. Harvesting and distribution planning output from the decision-support model.

4.2. Performance of the predict-then-optimize method

This section aims to test the performance of the proposed PO method and compare it with an empirical decision-making approach where farmers' price expectations are formulated based on prospect theory.

4.2.1. Formation of farmers' price expectations

Farmers typically make their production plans based on their estimations to future market conditions. For example, if farmers anticipate a significant rise in the price of vegetable A, they may increase its production. Conversely, they will reduce its production. According to Keynes' theory of irrational expectations, price expectations lack a reliable foundation and are vulnerable to emotional influences [38]. Given the volatility and irregularity of vegetable prices, farmers' expectations of future prices are often shaped by emotions. Letαrepresent the adjustment coefficient in price expectations, and p_t denote the expected price of farmers. p_{t-1} and p_{t-2} are the actual prices of period t-1 and t-2. Therefore, the irrational expectations can be expressed as $P_{t-1} + \alpha(P_{t-1} - P_{t-2})$. Based on this model, if $\alpha = 0$, the model becomes a static expectations model. In this case, producers typically expect the next period's market price to be the same as the current price, i.e., $P_t = P_{t-1}$. If the price of a particular vegetable increases from period t-2 to t-1, optimistic farmers might expect the price to continue rising. In this case, $\alpha = 1$, and the expected prices can be presented as$2P_{t-1} - P_{t-2}$. Pessimistic farmers however, may anticipate a sharp decline ($\alpha = -1$), and the expected prices can be denoted asP_{t-2}.

The above analysis shows that optimistic and pessimistic emotions lead to two opposing trends in farmers' price expectations for vegetables, as depicted in Fig. 6.

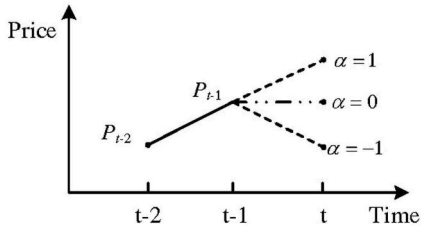


Fig. 6. The three types of expectation prices under irrational expectations.

In addition to Keynes' classical expectation theory, interviews with farmers reveal that many individuals rely on historical price data and use its average as their expectation for future market prices. In such cases, the price expectation model is represented by Eq. (13).

$$P_t = (P_{t-n} + P_{t-n-1} + \ldots + P_{t-1})/n, \quad n \in N* \tag{13}$$

4.2.2. Comparison process of decision efficiency between predicted and expected price

This section aims to verify whether the decision-making framework based on predicted prices is more effective than expected price in improving returns. To this end, we construct a comparison framework, which outlines two distinct decision-making paths: one based on data-driven predicted prices, and the other based on farmer expectations formed through emotional judgment or historical averaging. Although the visual paths in the figure appear similar, their key difference lies in the price formation mechanism, which ultimately leads to different profitability outcomes. The comparison is conducted by applying the decisions based on predicted prices and those based on expected prices to the same set of real market prices and evaluating the actual profits generated.

Table 4 reports the profit and cost results under different price reference models, while Fig. 7 provides an intuitive visual comparison across cultivation areas. Specifically, AP denotes actual prices, PP denotes predicted prices, ES1, ES2, and ES3 represent expected prices formed under optimistic, neutral, and pessimistic sentiments, respectively, and EH denotes the average historical price expectation. As shown in both Table 4 and Fig. 7(a), decisions based on PP consistently generate higher profits than those based on the four expectation-based price references across the 6 ha, 8 ha, and 10 ha cultivation areas. Although the profits under PP remain below those under AP, the corresponding profit gap (PG) is substantially smaller than that under the expectation-based scenarios, indicating that prediction-based decisions

Table 4
Comparison of unit profits and costs under predicted and expected prices.

Price	6 hm^2				8 hm^2				10 hm^2			
	Profit	PG	Cost	CG	Profit	PG	Cost	CG	Profit	PG	Cost	CG
AP	54,412	–	24,270	–	48,521	–	22,300	–	44,959	–	20,496	–
PP	47,293	-13%	30,349	+ 25%	44,501	-8%	26,592	+ 19%	42,358	-6%	23,918	+ 17%
ES1	39,040	-28%	23,356	-0.3%	36,086	-26%	20,558	-8%	33,672	-25%	18,759	-8.0%
ES2	39,914	-27%	23,547	-0.3%	34,696	-28%	23,687	+ 6%	32,602	-27%	22,869	+ 12%
ES3	40,385	-26%	24,020	-0.1%	34,446	-29%	25,679	+ 15%	32,559	-28%	23,007	+ 12%
EH	39,805	-27%	23,348	-4.0%	37,195	-23%	20,736	-7.0%	35,156	-22%	18,858	-8.0%

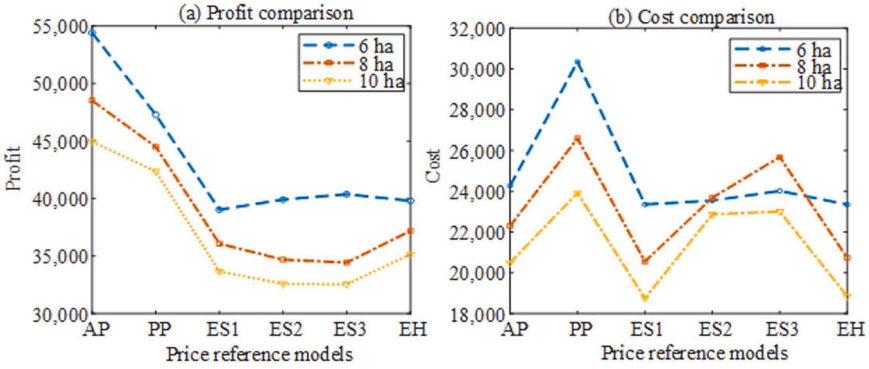


Fig. 7. Comparison of profit and cost under different price reference models and cultivation areas.

can achieve outcomes closer to those under actual prices. Fig. 7(b) further shows that decisions based on predicted prices are generally associated with higher costs. This is also reflected in the cost gap (CG) reported in Table 4, suggesting that the improved profitability of the PO approach is achieved through more active production and distribution adjustments, which may require greater financial support.

As further illustrated in Fig. 7, we also compare returns and costs under different price reference models across various cultivation areas. When the cultivation area is 6 ha, farmers with pessimistic sentiment achieve higher profits than those with optimistic sentiment. However, as the cultivation area expands, optimistic sentiment becomes more advantageous for increasing profitability. Using the average historical prices from the past four years is more beneficial for farmers than the classical expectation model, and this advantage grows as the cultivation area increases. The superiority of EH over the expectation model is due to the "first good, then bad" phenomenon in crop prices, where prices are high one year and low the next, alternating between two consecutive years.

The results confirm that decisions guided by predicted prices consistently outperform those based on farmer expectations, regardless of sentiment type or price reference model. This suggests that farm managers should incorporate predictive analytics into production planning to reduce reliance on subjective judgment and better adapt to market volatility. In practice, this may involve training programs in data interpretation, integrating prediction tools into farm management systems, and aligning crop and channel choices with forecasted market trends. Policymakers could facilitate adoption by improving farmers' access to reliable price forecasting services and promoting digital agriculture platforms that embed PO capabilities.

4.3. Sensitivity analysis of key factors

This section aims to examine how the key factors impact decision-making and profitability in agricultural production. We conduct sensitivity analysis on multi-crop cultivation, labor efficiency, available funds, and harvesting flexibility to provide a deeper understanding of their roles in optimizing farm operations.

Table 5
Profitability under different planting areas and crops.

Planting area (ha)	Profitability under different planting crops						
	T + P + C	T + P	T + C	P + C	T	P	C
2	1	1	0.77	0.84	0.74	0.84	0.81
4	1	0.98	0.77	0.88	0.67	0.84	0.76
6	1	0.96	0.72	0.88	0.65	0.81	0.76
8	1	0.95	0.70	0.88	0.66	0.80	0.78
10	1	0.94	0.70	0.88	0.68	0.81	0.81
Average	-	0.97	0.73	0.87	0.68	0.82	0.78

4.3.1. Multi-crop cultivation

This study uses tomatoes, peppers, and cucumbers as examples to analyze changes in profitability when farmers grow one, two, or three types of vegetables across different planting areas. The profitability values in Table 5 are based on the revenue from cultivating these three crops in various areas. The results show that for planting areas between 2 and 10 ha, cultivating three crops is more profitable than cultivating two. Additionally, the benefits of diversified cultivation become more apparent as the planting area increases.

The reason diversified cultivation boosts farmers' revenue is due to the varying price fluctuations of different crops. For instance, Fig. 2 in Section 3.2 shows that while the prices of peppers and cucumbers declined from December 2021 to February 2022, tomato prices increased. Thus, multi-crop cultivation helps farmers avoid over-reliance on a single crop, mitigates risks from natural disasters or market fluctuations, and enhances profitability.

Additionally, the enhanced profitability observed under diversified cultivation stems not only from mitigating price risks but also from the optimized utilization of labor, land, and other farm resources. Different crops have varying requirements regarding planting and harvesting schedules, resource inputs, and labor intensity. Tomatoes and cucumbers require intensive labor during harvesting periods but differ significantly in their peak labor demand weeks, allowing farms to allocate available labor resources more effectively and reduce idle labor time. Peppers, on the other hand, have relatively moderate labor demand distributed more evenly across the season, complementing the labor

patterns of tomatoes and cucumbers. Thus, multi-crop cultivation results in a more balanced workload, enhanced resource utilization efficiency, and overall productivity improvements.

Examining Table 5 more closely, it is clear that while cultivating tomatoes and peppers maintains high profitability close to the three-crop combination, combinations that exclude peppers generally yield lower returns. This pattern indicates that peppers serve as an important profitability stabilizer due to their distinct price trends and relatively stable demand. Furthermore, the economic advantage of diversification becomes increasingly evident at 8–10 ha, where the risk of single-crop market saturation and price volatility is greater. At 2–4 ha, profitability differences among crop combinations are less pronounced, suggesting that diversification becomes more crucial as farms expand their planting area and market exposure increases.

These results suggest that farm managers should prioritize diversified cultivation, especially when operating larger planting areas, to stabilize revenues and optimize labor allocation. Policymakers could further support diversification by providing targeted technical and market access programs.

4.3.2. Labor supply and efficiency

During harvest periods, there is a high demand for short-term labor. According to farmer interviews, approximately 50 man-weeks of labor are available per week. Based on the 2021 China Agricultural Cost and Income Data and field research, a single worker can manage 0.1 ha of tomatoes, 0.4 ha of peppers, and 0.085 ha of cucumbers daily in a typical greenhouse. Harvesting vegetables also requires significant labor: one worker can manually harvest 3000 kg of tomatoes, 1125 kg of peppers, and 1456 kg of cucumbers per week. Labor availability reflects the supply of labor, while per capita management area and harvest quantity indicate labor efficiency.

Using current labor supply and efficiency as a baseline, we examined changes in farm profitability by increasing labor supply and efficiency by 20%, up to five times. As shown in Fig. 8, both increasing the number of workers and improving labor efficiency significantly boost farm profitability. When labor shortages are no longer a constraint on vegetable production, farms can arrange their production plans more freely. Moreover, we found that the marginal benefits of increasing the number of workers decrease more sharply than those of improving labor efficiency.

In countries with aging populations, such as China, agricultural labor is relatively scarce, making it difficult for farms to expand their workforce. The findings indicate that enhancing labor efficiency, through training, mechanization, or automation, can generate sustained profitability gains even when the labor force is limited. Farmers should therefore view technology adoption and process optimization as strategic investments. Policymakers can support this transition by improving access to affordable agricultural machinery and relevant technical services.

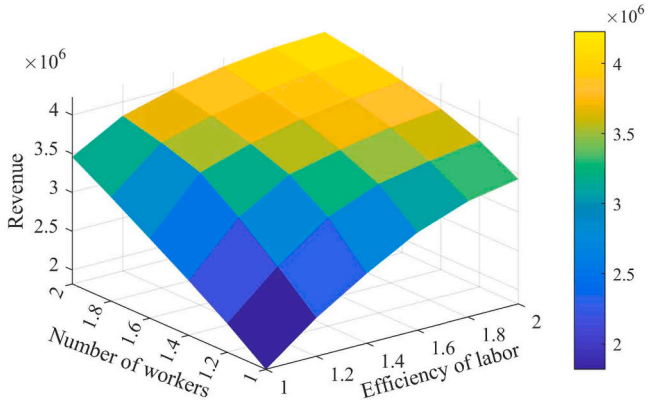


Fig. 8. The impact of the number of workers and labor efficiency on revenue.

4.3.3. Harvesting flexibility

Harvesting flexibility in crop production refers to the ability of farmers to decide whether to harvest immediately or delay harvest based on the physiological maturity and nutrient accumulation of crops. For comparison purposes, we assume a one-week shelf life for unharvested vegetables, allowing farmers to delay harvest by up to one week. To determine if harvesting flexibility can increase profits, we compared our proposed model (O-model) with a model that does not account for harvesting flexibility (B-model). As shown in Table 6, when the cultivable area exceeds 6 ha, harvesting flexibility leads to higher profits than immediate harvesting. This is because it enables farms to supply more products to the market when prices are higher.

To further investigate the interaction between labor input and harvesting flexibility, we extended the analysis by introducing different levels of labor availability and efficiency, denoted by L. In addition to the baseline case (L=1), we analyzed scenarios with moderate (L=1.5) and high (L=2) labor capacity. As shown in Table 6, increasing L consistently leads to higher profits across both models, but the relative benefit of harvesting flexibility remains significant. For instance, at L= 1.5, the profit gain from adopting flexible harvesting reaches up to 34.7% in the 10-hectare scenario. This suggests that even when labor constraints are relaxed, harvesting flexibility continues to play an important role in capturing market value by enabling better timing of product distribution. The combined impact of enhanced labor efficiency and harvesting flexibility illustrates the value of jointly improving operational capacity and decision-making adaptability.

The analysis demonstrates that harvesting flexibility enables farms to better align supply with market demand and capture higher prices, even under labor constraints. Managers should integrate flexible harvesting strategies into their operational planning, while policymakers could promote their adoption by investing in storage facilities and post-harvest handling technologies.

4.3.4. Financial resources

We test the impact of available funds on the production and management of multi-crop cultivation farms, focusing on an 8 hectare planting area. By adjusting the available funds per hectare from 250 K to 300 K, we observe changes in farm operations.

As shown in Table 7, increasing the available funds from RMB 25,000/ha to RMB 30,000/ha leads to a noticeable shift in the crop sales structure. At lower funding levels, sales are heavily skewed toward peppers; for instance, when the available funds are 25,000 RMB/ha, peppers account for 79.07% of total crop sales, whereas tomatoes and cucumbers are either underrepresented or not sold at all. As funding increases, the proportion of tomatoes rises steadily, and cucumbers are introduced into the sales mix starting at 28.75 K/ha. This trend reflects the fact that tomatoes and cucumbers, while potentially more profitable through high-value channels, also require higher upfront investment in planting, packaging, and logistics. The more balanced crop mix at higher funding levels demonstrates that adequate financial resources enable farmers to diversify production more effectively and tap into a wider range of market opportunities. Furthermore, the steady increase in per-hectare profits—from RMB 281.74 K to RMB 326.69 K—confirms that capital constraints are a significant limiting factor in optimizing planting and distribution strategies. These findings underscore the importance of financial support mechanisms, such as credit access and subsidies, especially for smallholder farms aiming to adopt multi-crop and multi-channel strategies.

Increased funds also influence the selection of sales channels. As shown in Table 8, when available funds rise from 250 K/ha to 300 K/ha, the proportion of vegetable sales through higher-priced channels like fresh e-commerce and retail increases. Adequate funding gives farms greater flexibility in choosing sales channels, allowing them to sell products at higher prices. Therefore, sufficient funds are essential for

Table 6
Profitability of flexible harvesting under different land and labor conditions.

L		6 ha		8 ha		10 ha	
		B-model	Our model	B-model	O-model	B-model	O-model
1	Profit	2588,986	2947,941	3178,127	3566,147	3458,401	3851,050
	Profit gap	+ 13.86%		+ 12.21%		+ 11.35%	
1.5	Profit	2671,004	3293,528	3175,781	3913,325	3268,330	4403,283
	Profit gap	+ 23.30%		+ 23.22%		+ 34.7%	
2	Profit	2781,228	3428,066	3459,132	4224,517	3993,209	4865,028
	Profit gap	+ 23.26%		+ 22.12%		+ 21.83%	

Table 7
The impact of available funds on the proportion of crop sales.

Available Funds (RMB/ha)	Sales proportion of each crop (%)			Profits/ha
	T	P	C	
250.00 K	20.93	79.07	-	281.74 K
260.25 K	22.30	77.70	-	296.63 K
270.50 K	27.01	72.99	-	310.86 K
280.75 K	26.64	66.58	6.78	318.95 K
300.00 K	24.70	61.38	13.92	326.69 K

Table 8
The impact of available funds on distribution channel sales.

Available Funds (RMB/ha)	Crop	Proportion of sales (%)			
		Contract farming	Fresh e-commerce	Wholesale	Retail
250.00 K	T	30.00	53.90	-	16.10
	P	45.00	16.07	7.68	31.25
	C	-	-	-	-
Total proportion		41.86	23.99	6.07	28.08
270.50 K	T	30.00	41.79	-	28.21
	P	44.18	15.46	8.83	31.53
	C	-	-	-	-
Total proportion		40.35	22.57	6.45	30.63
300.00 K	T	30.00	42.70	-	27.30
	P	44.83	17.18	5.37	32.62
	C	30.00	28.00	-	42.00
Total proportion		39.10	24.99	3.29	32.61

maximizing the income potential of multi-crop cultivation and diversified sales channels. The results highlight that adequate financial resources allow for more profitable crop combinations and greater participation in high-value sales channels. Farmers should align production strategies with their capital availability, while policymakers and rural financial institutions can enhance agricultural profitability by improving credit access, expanding loan limits, and designing tailored financial products for diversified farming.

The sensitivity analysis shows that farm profitability is influenced by a combination of factors, including production diversity, labor conditions, harvesting flexibility, and financial availability, but these factors operate through different mechanisms. Diversification primarily enhances profitability by mitigating market risk and improving the utilization of land and labor resources. Increased labor efficiency yields more sustainable marginal benefits than simply increasing labor supply, highlighting the importance of mechanization and operational upgrades. Harvest flexibility improves a farm's ability to align supply with market conditions, while financial resources affect both the feasible scale of production and the choice of crop mix and sales channels. These results suggest that improving farm performance under price uncertainty requires not a single intervention but a coordinated strategy combining production diversification, labor-saving technologies, flexible operations, and financial support.

4.4. Managerial implications for the application of the proposed approach

The findings of this study provide several managerial implications for the proposed PO approach in fresh agricultural production systems. For family farms, this method can serve as a practical decision support tool for coordinating crop selection, planting plans, harvest timing, and channel allocation under price uncertainty. This is particularly important for farms adopting multi-crop planting, flexible harvesting, and multi-channel distribution, as these operational characteristics create strong interdependencies among production, harvesting, and distribution decisions. Compared with experience-based planning, using predicted prices helps reduce reliance on subjective judgment and improves the quality of planning decisions.

For agricultural cooperatives, agricultural service agencies, and digital agriculture platforms, this method can be integrated into decision support systems to generate more structured planning recommendations. Its value is especially evident in fresh produce management, as market timing, perishability, and channel differences collectively impact profitability. By linking upstream planting and harvesting decisions with downstream sales opportunities, this method can support better coordination across the production and distribution chain.

Sensitivity analysis also provides insights for agricultural support policies and related services. The positive impact of labor efficiency highlights the importance of mechanization, worker training, and labor-saving technologies. The profitability gains associated with harvest flexibility demonstrate the significant value of post-harvest support and operational arrangements that support more flexible harvesting arrangements. Furthermore, the substantial impact of financial availability on crop mix and channel choice suggests that access to credit and tailored financial support can help farms better capitalize on high-value market opportunities.

5. Conclusion

In this paper, we developed a comprehensive PDP model for family farms based on the predict-then-optimize decision framework. The model integrates key operational aspects, including planting, harvesting, and distribution, within the context of multi-crop cultivation, flexible harvesting, and multi-channel sales, while leveraging predicted prices to inform decision-making. The case study demonstrates the model's practical applicability and provides valuable management insights through numerical experiments.

Our results confirm that the PO-based approach significantly outperforms traditional experiential methods in enhancing farm profitability. Specifically, decisions guided by predicted prices lead to consistently higher profits compared to those based on farmers' expected prices. This highlights the effectiveness of using data-driven predictions for optimizing decisions on crop types, planting schedules, cultivation areas, harvest timing, and sales channels. Additionally, the incorporation of multi-crop cultivation, multi-channel distribution, and flexible harvesting further improves farm profitability by allowing farmers to better respond to price fluctuations and optimize resource allocation.

The findings of this study also provide practical implications for the

application of the proposed approach. For family farms, the model can serve as a decision-support tool for coordinating crop, harvest, and channel decisions under price uncertainty. For agricultural service providers and policymakers, the results highlight the importance of predictive analytics, labor-saving technologies, harvesting flexibility, and financial support in improving farm profitability.

Despite the promising results, the proposed decision framework has certain limitations that provide potential directions for future research. First, the prediction and decision-making processes in our model operate independently, lacking a dynamic feedback loop between the two. Establishing a direct linkage between predictions and decision outcomes could improve the overall robustness of the framework. Second, the planning horizon in this study is limited to five months, aligned with the lifecycle of a single crop, which restricts the exploration of crop rotation strategies involving a wider variety of crops. Extending the model's planning scope to accommodate longer-term crop rotation scenarios could enhance its applicability. Finally, while demand is treated as a deterministic parameter in this study, real-world demand is often subject to fluctuations and interacts with prices. Future iterations of the model should incorporate joint predictions of demand and prices to provide a more comprehensive decision-making tool.

In conclusion, our study demonstrates the potential of the predict-then-optimize framework to revolutionize decision-making in agricultural production by providing more accurate insights. Our findings suggest that incorporating predictive analytics into farm management can help farmers mitigate risks associated with price volatility, improve operational efficiency, and ultimately boost profitability. Future research should focus on refining the model to address its current limitations and further exploring the integration of dynamic market factors, which could provide even greater support for farmers responding complex agricultural environments.

CRediT authorship contribution statement

Xinxin Liu: Visualization, Investigation, Conceptualization. **Na Lin:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Junhu Ruan:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Conceptualization. **Yuanwei Sun:** Visualization, Software, Methodology. **Jiliang Han:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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